Exercitii:

① Aflati extremele locale pentru:

a) $f(x,y)=x^3+y^3+6xy+24$

$$\begin{cases} \frac{\partial f}{\partial x}=3x^2+6y=0 \\ \frac{\partial f}{\partial y}=3y^2+6x=0 \end{cases} \Rightarrow \begin{cases} y=\frac{-x^2}{2} \\ 3\cdot\frac{x^4}{4}+6x=0 \Rightarrow \frac{x^4}{4}+2x=0 \Rightarrow \end{cases}$$

$\Rightarrow x^4+8x=0 \Rightarrow x(x^3+8)=0 \Rightarrow x(x+2)(x^2-2x+4)=0$

$\Delta=4-4\cdot1\cdot4<0$ nu are sol. reale

$x_1=0 \Rightarrow y_1=0 \Rightarrow M_1(0;0)$

$x_2=-2 \Rightarrow y_2=-2 \Rightarrow M_2(-2;-2)$

$$\frac{\partial^2 f}{\partial x^2}=6x \qquad H(x,y)=\begin{pmatrix} 6x & 6 \\ 6 & 6y \end{pmatrix}$$

$$\frac{\partial^2 f}{\partial y^2}=6y$$

$$\frac{\partial^2 f}{\partial x\partial y}=6$$

$$H(0;0)=\begin{pmatrix} 0 & 6 \\ 6 & 0 \end{pmatrix} \qquad \det(H-\lambda J)=0 \Rightarrow \begin{vmatrix} -\lambda & 6 \\ 6 & -\lambda \end{vmatrix}=0$$

$$\Rightarrow \lambda^2-36=0 \Rightarrow \lambda=\pm 6$$

$\Rightarrow M_1(0;0)$ nu e punct de extreem

$$H(-2;-2)=\begin{pmatrix} -12 & 6 \\ 6 & -12 \end{pmatrix} \qquad \Delta_1=-12<0$$

$$\Delta_2=6\cdot6\cdot\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix}=36\cdot(4-1)>0$$

$(-1)\cdot\Delta_1 > 0$
$(-1)^2\cdot\Delta_2 > 0$
$\Rightarrow$ $M_2(-2;-2)$ pct. de maxim local

b) $f(x,y) = (x+y)e^{-(x^2+y^2)}$

$$\begin{cases} \frac{\partial f}{\partial x} = e^{-(x^2+y^2)} + (x+y)\cdot e^{-(x^2+y^2)}(-2x) = 0 \\ \frac{\partial f}{\partial y} = e^{-(x^2+y^2)} + (x+y)\cdot e^{-(x^2+y^2)}(-2y) = 0 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} e^{-(x^2+y^2)}(1-2x^2-2xy) = 0 \\ e^{-(x^2+y^2)}(1-2xy-2y^2) = 0 \end{cases} \Rightarrow \begin{cases} 1-2x^2-2xy = 0 \\ 1-2xy-2y^2 = 0 \end{cases}$$

$\Rightarrow 2y^2-2x^2 = 0 \Rightarrow (y-x)(y+x) = 0$

I $x = y \Rightarrow 1-2x^2-2x^2 = 0 \Rightarrow 1-4x^2 = 0 \Rightarrow 4x^2 = 1 \Rightarrow$
$\Rightarrow x^2 = \frac{1}{4} \Rightarrow x = \pm\frac{1}{2}$ $\quad M_1\left(\frac{1}{2};\frac{1}{2}\right)$
$M_2\left(-\frac{1}{2};-\frac{1}{2}\right)$

II $x = -y \Rightarrow 1-2y^2+2y^2 = 0 \Rightarrow 1 = 0$ Fals

$$\frac{\partial^2 f}{\partial x^2} = e^{-(x^2+y^2)}(-2x)(1-2x^2-2xy) + e^{-(x^2+y^2)}(-4x-2y)$$
$$= e^{-(x^2+y^2)}(-6x+4x^3+4x^2y-2y)$$

$$\frac{\partial^2 f}{\partial y^2} = e^{-(x^2+y^2)}(-2y)(1-2xy-2y^2) + e^{-(x^2+y^2)}(-2x-4y)$$
$$= e^{-(x^2+y^2)}(-6y+4xy^2+4y^3-2x)$$

$$\frac{\partial f}{\partial x \partial y} = e^{-(x^2+y^2)}(-2x)(1-2xy-2y^2) + e^{-(x^2+y^2)}(-2y)$$

$$= e^{-(x^2+y^2)}(-2x + 4x^2y + 4xy^2 - 2y)$$

$$H\left(\frac{1}{2};\frac{1}{2}\right) = \begin{pmatrix} -3\cdot e^{-1/2} & -e^{-1/2} \\ -e^{-1/2} & -3\cdot e^{-1/2} \end{pmatrix}$$

$\Delta_1 < 0$ $\quad (-1)^1 \cdot \Delta_1 > 0$

$\Delta_2 = 9e^{-1} - e^{-1} > 0 \Rightarrow (-1)^2 \cdot \Delta_2 > 0$ $\Rightarrow$ $\left(\frac{1}{2};\frac{1}{2}\right)$ punct de maxim local

$$H\left(-\frac{1}{2};-\frac{1}{2}\right) = \begin{pmatrix} e^{-1/2}\cdot 3 & e^{-1/2} \\ e^{-1/2} & e^{-1/2}\cdot 3 \end{pmatrix}$$

$\Delta_2 = 9e^{-1} - e^{-1} \Rightarrow \Delta_2 > 0$, $\Delta_1 > 0$ $\Rightarrow$ $M_1\left(-\frac{1}{2};\frac{-1}{2}\right)$ pct. de minim local

c) $f(x,y) = x^2 y e^{2x+3y}$

$$\frac{\partial f}{\partial x} = 2xy e^{2x+3y} + x^2 y \cdot e^{2x+3y}\cdot 2 = y e^{2x+3y}(2x + 2x^2)$$

$$\frac{\partial f}{\partial y} = x^2 e^{2x+3y} + x^2 y e^{2x+3y}\cdot 3 = x^2 e^{2x+3y}(1+3y)$$

$$\begin{cases} 2xy(x+1) = 0 \\ x^2(1+3y) = 0 \end{cases}$$

I $x=-1 \Rightarrow y=-\frac{1}{3} \Rightarrow M_1\left(-1;-\frac{1}{3}\right)$

II $x=0 \Rightarrow y\in\mathbb{R} \Rightarrow M_2(0;y)$

$$\frac{\partial^2 f}{\partial x^2} = y\cdot e^{2x+3y}\cdot 2(2x+2x^2) + ye^{2x+3y}(2+4x)$$

$$= 2ye^{2x+3y}(1+4x+2x^2)$$

$$\frac{\partial^2 f}{\partial y^2} = x^2\cdot e^{2x+3y}\cdot 3(1+3y) + x^2e^{2x+3y}\cdot 3$$

$$= 3x^2e^{2x+3y}(2+3y)$$

$$\frac{\partial^2 f}{\partial x\partial y} = 2xe^{2x+3y}(1+3y) + x^2e^{2x+3y}\cdot 2(1+3y)$$

$$= 2xe^{2x+3y}(1+3y)(1+x)$$

$$H\left(-1;-\frac{1}{3}\right) = \begin{pmatrix} 2/3e^3 & 0 \\ 0 & 3/e^3 \end{pmatrix}$$

$\Delta_1>0$, $\Delta_2>0$ $\Rightarrow$ $\left(-1;-\frac{1}{3}\right)$ pct. de minim local

$$H(0;y) = \begin{pmatrix} 2ye^{3y} & 0 \\ 0 & 0 \end{pmatrix}$$

$$f(x,y)-f(0,y) = x^2ye^{2x+3y} - 0 = x^2ye^{2x+3y}$$

$$f(x,y)-f(0,y) = \begin{cases} +, y>0 & \Rightarrow (0;y) \text{ minim loc.} \\ -, y<0 & \Rightarrow (0;y) \text{ maxim local} \end{cases}$$

pt. $y=0 \Rightarrow f(x,y)-f(0,0)=x^2 y e^{2x+3y}$ care nu păstrează semn constant în vecinătatea lui $(0;0)$

$\Rightarrow$ $(0;0)$ nu e extrem local

d) $f(x,y,z)=\frac{1}{y}+\frac{y}{z}+\frac{z}{x}+x+1 \qquad x,y,z\neq 0$

$$\begin{cases} \frac{\partial f}{\partial x}=z\cdot(-1)\cdot\frac{1}{x^2}+1=0 \\ \frac{\partial f}{\partial y}=-\frac{1}{y^2}+\frac{1}{z}=0 \\ \frac{\partial f}{\partial z}=y\cdot\left(-\frac{1}{z^2}\right)+\frac{1}{x}=0 \end{cases} \Rightarrow \begin{cases} \frac{z}{x^2}=1 \Rightarrow z=x^2 \\ \frac{1}{z}=\frac{1}{y^2} \Rightarrow z=y^2 \\ \frac{1}{x}=\frac{y}{z^2} \Rightarrow z^2=xy \end{cases}$$

$z^2=xy \Rightarrow y^4=xy \Rightarrow y^4-xy=0 \Rightarrow y(y^3-x)=0 \Rightarrow y^3=x$
pt. că $y\neq 0$

$z=x^2 \Rightarrow z=y^6$
$z=y^2$ $\Rightarrow y^2=y^6 \Rightarrow y^4=1 \Rightarrow y=\pm 1$

$y_1=1 \Rightarrow x_1=1 \Rightarrow z_1=1 \Rightarrow M_1(1;1;1)$

$y_2=-1 \Rightarrow x_2=-1 \Rightarrow z_2=1 \Rightarrow M_2(-1;-1;1)$

$\frac{\partial^2 f}{\partial x^2}=-z\cdot(-2)\cdot\frac{1}{x^3}=\frac{2z}{x^3}$ $\qquad \frac{\partial^2 f}{\partial x\partial y}=0$

$\frac{\partial^2 f}{\partial y^2}=-(-2)\cdot\frac{1}{y^3}=\frac{2}{y^3}$ $\qquad \frac{\partial^2 f}{\partial x\partial z}=-\frac{1}{x^2}$

$\frac{\partial^2 f}{\partial z^2}=y\cdot(-1)\cdot(-2)\cdot\frac{1}{z^3}=\frac{2y}{z^3}$ $\qquad \frac{\partial^2 f}{\partial y\partial z}=-\frac{1}{z^2}$

$$Hf(1;1;1)=\begin{pmatrix}2&0&-1\\0&2&-1\\-1&-1&2\end{pmatrix}$$

$\Delta_1=2\Rightarrow\Delta_1>0$

$\Delta_2=4\Rightarrow\Delta_2>0$

$\Rightarrow$

$$\Delta_3=\begin{vmatrix}2&0&-1\\0&2&-1\\-1&-1&2\end{vmatrix}=8-2-2=4\Rightarrow\Delta_3>0$$

$\Rightarrow$ $M_1(1;1;1)$ punct de minim

$$Hf(-1;-1;1)=\begin{pmatrix}-2&0&-1\\0&-2&-1\\-1&-1&-2\end{pmatrix}$$

$\Delta_1=-2\Rightarrow(-1)\Delta_1>0$

$\Delta_2=4\Rightarrow(-1)^2\Delta_2>0$

$$\Delta_3=\begin{vmatrix}-2&0&-1\\0&-2&-1\\-1&-1&-2\end{vmatrix}=-8+2+2=-4$$

$\Delta_3(-1)^3>0$

$\Rightarrow$ $M_2(-1;-1;1)$ e punct de maxim local

c) $f(x,y,z)=x+\frac{y^2}{4x}+\frac{z^2}{y}+\frac{2}{z}$ $\quad x,y,z\neq0$

$$\begin{cases}\frac{\partial f}{\partial x}=1+\frac{y^2}{4}\cdot\left(-\frac{1}{x^2}\right)=0\\ \frac{\partial f}{\partial y}=\frac{2y}{4x}+z^2\cdot\left(-\frac{1}{y^2}\right)=0\\ \frac{\partial f}{\partial z}=\frac{2z}{y}+2\cdot\left(-\frac{1}{z^2}\right)=0\end{cases}\Rightarrow\begin{cases}\frac{y^2}{x^2}=4\Rightarrow y^2=4x^2\\ \frac{y}{2x}=\frac{z^2}{y^2}\Rightarrow y^3=2x\cdot z^2\Rightarrow x=\frac{y^3}{2z^2}\\ \frac{z}{y}=\frac{1}{z^2}\Rightarrow y=z^3\end{cases}$$

$$\Rightarrow y^2=4\cdot\left(\frac{y^3}{2z^2}\right)^2\Rightarrow z^6=4\cdot\frac{y^6}{4z^4}\Rightarrow z^{10}=z^{18}\Rightarrow z=\pm1$$

$z_1 = 1 \Rightarrow y_1 = 1 \Rightarrow x_1 = \frac{1}{2} \quad \Rightarrow M_1\left(\frac{1}{2}; 1; 1\right)$

$z_2 = -1 \Rightarrow y_2 = -1 \Rightarrow x_2 = \frac{-1}{2} \quad \Rightarrow M_2\left(-\frac{1}{2}; -1; -1\right)$

$$\frac{\partial^2 f}{\partial x^2} = -\frac{y^2}{4} \cdot (-2) \cdot \frac{1}{x^3} = \frac{y^2}{2x^3}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{1}{2x} + (-z^2) \cdot (-2) \cdot \frac{1}{y^3} = \frac{1}{2x} + \frac{2z^2}{y^3}$$

$$\frac{\partial^2 f}{\partial z^2} = \frac{2}{y} + (-2) \cdot (-2) \cdot \frac{1}{z^3} = \frac{2}{y} + \frac{4}{z^3}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{y}{2} \cdot (-1) \cdot \frac{1}{x^2} = \frac{-y}{2x^2}$$

$$\frac{\partial^2 f}{\partial x \partial z} = 0$$

$$\frac{\partial^2 f}{\partial y \partial z} = 2z \cdot (-1) \cdot \frac{1}{y^2} = \frac{-2z}{y^2}$$

$$Hf\left(\frac{1}{2}; 1; 1\right) = \begin{pmatrix} 4 & -2 & 0 \\ -2 & 3 & -2 \\ 0 & -2 & 6 \end{pmatrix}$$

$\Delta_1 = 4 > 0$

$\Delta_2 = 12 - 4 = 8 > 0$

$\Delta_3 = 12 \cdot 6 - 16 - 4 \cdot 6 = 32 > 0$

$\Rightarrow$ $\left(\frac{1}{2}; 1; 1\right)$ pct. de minim local

$$Hf\left(-\frac{1}{2}; -1; -1\right) = \begin{pmatrix} -4 & 2 & 0 \\ 2 & -3 & 2 \\ 0 & 2 & -6 \end{pmatrix}$$

$\Delta_1 = -4 < 0 \Rightarrow (-1)^1 \cdot \Delta_1 > 0$

$\Delta_2 = 12 - 4 = 8 \Rightarrow (-1)^2 \Delta_2 > 0$

$\Delta_3 = -12 \cdot 6 + 16 + 6 \cdot 4 = -32 < 0 \Rightarrow (-1)^3 \Delta_3 > 0$

$\Rightarrow \left(-\frac{1}{2}; -1; -1\right)$ pct. de maxim local

f) $f(x,y) = x^4 + y^4 - x^2 - y^2 - xy$

$$\begin{cases} \frac{\partial f}{\partial x} = 4x^3 - 2x - y = 0 \\ \frac{\partial f}{\partial y} = 4y^3 - 2y - x = 0 \end{cases} \Rightarrow \begin{array}{l} 4(x^3 - y^3) - 2(x-y) - (y-x) = 0 \\ 4(x-y)(x^2+xy+y^2) - (x-y) = 0 \\ (x-y)\left[4(x^2+xy+y^2) - 1\right] = 0 \end{array}$$

I $x = y \Rightarrow 4x^3 - 2x - x = 0 \Rightarrow 4x^3 - 3x = 0 \Rightarrow x(4x^2 - 3) = 0$

$\Rightarrow$ I $x = 0 \Rightarrow y = 0 \quad M_1(0;0)$

II $4x^2 - 3 = 0 \Rightarrow x^2 = \frac{3}{4} \Rightarrow x = \pm \frac{\sqrt{3}}{2} \Rightarrow M_2\left(\frac{\sqrt{3}}{2}; \frac{\sqrt{3}}{2}\right)$

$M_3\left(-\frac{\sqrt{3}}{2}; -\frac{\sqrt{3}}{2}\right)$

II $4x^2 + 4xy + 4y^2 - 1 = 0 \Rightarrow \begin{cases} 4x^2 + 4xy + 4y^2 = 1 \\ 4x^3 - 2x - y = 0 \end{cases} \Rightarrow$

$\Rightarrow \begin{cases} 4x^2 + 4x(4x^3 - 2x) + 4(4x^3 - 2x)^2 = 1 \\ y = 4x^3 - 2x \end{cases} \Rightarrow$

$\Rightarrow 4x^2 + 16x^4 - 8x^2 + 4(16x^6 - 16x^4 + 4x^2) = 1$

$\Rightarrow 16x^4 - 4x^2 + 4 \cdot 16x^6 - 4 \cdot 16x^4 + 16x^2 = 1$

$\Rightarrow 4 \cdot 16x^6 - 3 \cdot 16x^4 + 12x^2 = 1 \Rightarrow (4x^2)^3 - 4 \cdot 12x^4 + 12x^2 - 1 = 0$

$\Rightarrow (4x^2)^3 - 1 + 12x^2(1 - 4x^2) = 0 \Rightarrow$

$(4x^2-1)\left[(4x^2)^2+(4x^2)+1\right]-12x^2(4x^2-1)=0$

$(4x^2-1)\left[16x^4+4x^2+1-12x^2\right]=0 \Rightarrow (4x^2-1)(16x^4-8x^2+1)=0$

$\Rightarrow (4x^2-1)^3=0 \Rightarrow 4x^2-1=0 \Rightarrow x^2=\frac{1}{4} \Rightarrow x=\pm\frac{1}{2}$

$x=\frac{1}{2} \Rightarrow y=4\cdot\frac{1}{8}-1=-\frac{1}{2} \Rightarrow M_4\left(\frac{1}{2};-\frac{1}{2}\right)$

$x=-\frac{1}{2} \Rightarrow y=4\cdot-\frac{1}{8}+1=1-\frac{1}{2}=\frac{1}{2} \Rightarrow M_5\left(-\frac{1}{2};\frac{1}{2}\right)$

$\frac{\partial^2 f}{\partial x^2}=12x^2-2 \qquad \frac{\partial^2 f}{\partial y^2}=12y^2-2 \qquad \frac{\partial^2 f}{\partial x\partial y}=-1$

$H(0;0)=\begin{pmatrix}-2 & -1\\ -1 & -2\end{pmatrix}$ $\quad \Delta_1\cdot(-1)>0$, $\Delta_2=4-1=3$, $\Delta_2\cdot(-1)^2>0 \Rightarrow$

$\Rightarrow$ $M_1(0;0)$ punct de maxim local

$H\left(\frac{\pm\sqrt{3}}{2};\frac{\pm\sqrt{3}}{2}\right)=\begin{pmatrix}+ & -1\\ -1 & +\end{pmatrix}$ $\quad \Delta_1>0$, $\Delta_2>0 \Rightarrow$ $M_2\left(\frac{\sqrt{3}}{2};\frac{\sqrt{3}}{2}\right)$ și $M_3\left(-\frac{\sqrt{3}}{2};-\frac{\sqrt{3}}{2}\right)$

pct. de minim local

$H\left(\frac{\pm 1}{2};\frac{\mp 1}{2}\right)=\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}$ $\quad \Delta_1>0$, $\Delta_2=0 \Rightarrow$ nu merge această metodă

$\det(H-\lambda J)=\begin{vmatrix}1-\lambda & -1\\ -1 & 1-\lambda\end{vmatrix}=(1-\lambda)^2-1=1-2\lambda-\lambda^2-1$

$=-\lambda(2-\lambda) \Rightarrow$ nu merge această metodă

$f(x,y)-f\left(\frac{1}{2};-\frac{1}{2}\right)=\frac{1}{1!}\left[(x-\frac{1}{2})\cdot 0+(y+\frac{1}{2})\cdot 0\right]+\frac{1}{2!}\left[(x-\frac{1}{2})^2\cdot 1\right.$

$\left.+2\cdot(x-\frac{1}{2})(y+\frac{1}{2})\cdot(-1)+(y+\frac{1}{2})^2\cdot 1\right]=\frac{1}{2!}\left[(x-\frac{1}{2})-(y+\frac{1}{2})\right]^2>0$

$\Rightarrow$ $M_4\left(\frac{1}{2};-\frac{1}{2}\right)$ pct. de minim

$f(x,y) - f\left(-\frac{1}{2};\frac{1}{2}\right) = \frac{1}{1!}\left[\left(x+\frac{1}{2}\right)\cdot 0 + \left(y-\frac{1}{2}\right)\cdot 0\right] + \frac{1}{2!}\left[\left(x+\frac{1}{2}\right)^2\cdot 1 + \right.$

$\left. + 2\left(x+\frac{1}{2}\right)\left(y-\frac{1}{2}\right)\cdot(-1) + \left(y-\frac{1}{2}\right)^2\cdot 1\right] = \frac{1}{2!}\left[\left(x+\frac{1}{2}\right)-\left(y+\frac{1}{2}\right)\right]^2 > 0$

$\Rightarrow M_5\left(-\frac{1}{2};\frac{1}{2}\right)$ pct. de minim

② Determinați valorile extreme ale funcției

a) $f(x,y) = 2(x^2+y^2+x)$ pe mulțimea $K: x^2+y^2 \leq 1$

K mulțime compactă $\overset{\text{Th.W.}}{\Longrightarrow}$ f mărginită și își atinge marginile

f continuă

$\Rightarrow \exists \sup_K f$ și $\inf_K f$

1) Pe $\mathring{K}: x^2+y^2<1$ ($\mathring{K}$ interiorul mult)

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \Rightarrow \begin{cases} 4x+2=0 \Rightarrow x=-\frac{1}{2} \\ 4y=0 \Rightarrow y=0 \end{cases}$$

$\left(-\frac{1}{2};0\right) \in \mathring{K}$

pt. că $\frac{1}{4}+0<1$

2) Pe Frontiera $K: x^2+y^2=1$ $\quad g(x,y) = x^2+y^2-1$

$F(x,y,\lambda) = f(x+y) + \lambda g(x,y) = 2(x^2+y^2+x) + \lambda(x^2+y^2-1)$

$$\begin{cases} \frac{\partial F}{\partial x} = 0 \\ \frac{\partial F}{\partial y} = 0 \\ g(x,y) = 0 \end{cases} \Rightarrow \begin{cases} 4x+2+2\lambda x = 0 \\ 4y+2\lambda y = 0 \\ x^2+y^2-1=0 \end{cases}$$

$\Rightarrow 2y(2+\lambda)=0$

I $y=0 \Rightarrow x=\pm 1$

II $\lambda=-2 \Rightarrow 4x+2-4x=0 \Rightarrow \nexists$

$\Rightarrow$ avem punctele $(1;0)$ $(-1;0)$

$f(1;0) = 2(1+0+1) = 4 \longrightarrow \sup_K f = (1;0)$

$f(-1;0) = 2(1+0-1) = 0$

$f\left(-\frac{1}{2};0\right) = 2\left(\frac{1}{4}+0-\frac{1}{2}\right) = 2\cdot\left(-\frac{1}{4}\right) = -\frac{1}{2} \Rightarrow \inf_K f = \left(-\frac{1}{2};0\right)$

b) $f(x,y) = 5 - x^2 - (y-2)^2$ pe multimea $K: x^2+y^2 \leq 2$

K multime compactă, f continuă $\overset{\text{Th.W.}}{\Longrightarrow}$ f mărginită și își atinge marginile

$\Rightarrow \exists \sup_K f$ și $\inf_K f$

1) Pe $\mathring{K}: x^2+y^2 < 2$

$$\begin{cases} \frac{\partial f}{\partial x} = -2x = 0 \\ \frac{\partial f}{\partial y} = 2(y-2) = 0 \end{cases} \Rightarrow (0;2) \notin \mathring{K} \text{ pt. că } 4 < 2 \text{ fals}$$

2) Pe frontiera $K: x^2+y^2 = 2$ $\quad g(x,y) = x^2+y^2-2$

$F(x,y,\lambda) = 5 - x^2 - (y-2)^2 + \lambda g(x,y)$

$$\begin{cases} \frac{\partial F}{\partial x} = -2x + 2\lambda x = 0 \\ \frac{\partial F}{\partial y} = 2(y-2) + 2\lambda y = 0 \\ g(x,y) = 0 \end{cases} \Rightarrow \begin{cases} 2x(\lambda-1) = 0 \\ 2y(1-\lambda) = 4 \\ x^2+y^2 = 2 \end{cases}$$

I $x = 0 \Rightarrow y^2 = 2 \Rightarrow y = \pm\sqrt{2} \Rightarrow M_1(0;\sqrt{2})$ $M_2(0;-\sqrt{2})$ /

II $\lambda - 1 = 0 \Rightarrow \lambda = 1 \Rightarrow 2y \cdot 0 = 4$ ✗ $\Rightarrow$ Fals $\Rightarrow$

$\Rightarrow$ avem punctele $(0;\sqrt{2}); (0;-\sqrt{2})$

$f(0;\sqrt{2}) = 5 - 0 - (\sqrt{2}-2)^2 = 5 - (2 - 4\sqrt{2} + 4) = 5 - 6 + 4\sqrt{2}$

$= 4\sqrt{2} - 1 \Rightarrow \sup_K f = 4\sqrt{2} - 1$

$f(0;-\sqrt{2}) = 5 - 0 - (-\sqrt{2}-2)^2 = 5 - (2 + 4\sqrt{2} + 4) = 5 - 6 - 4\sqrt{2}$

$= -1 - 4\sqrt{2} \Rightarrow \inf_K f = -1 - 4\sqrt{2}$

c) $f(x,y) = xy$ pe multimea $K: x^2 + 8y^2 \leq 1$

K multime compactă $\overset{\text{Th. W.}}{\Longrightarrow}$ f mărginită și își atinge marginile

f continuă

$\Rightarrow \exists \sup_K f$ și $\inf_K f$

1) Pe $\mathring{K}: x^2 + 8y^2 < 1$

$$\begin{cases} \frac{\partial f}{\partial x} = y = 0 \\ \frac{\partial f}{\partial y} = x = 0 \end{cases} \Rightarrow (0;0) \in \mathring{K} \text{ pt că } 0^2 + 8 \cdot 0^2 < 1$$

2) Pe frontiera $K: x^2 + 8y^2 = 1 \quad g(x,y) = x^2 + 8y^2 - 1$

$F(x,y,\lambda) = xy + \lambda g(x,y)$

$$\begin{cases} \frac{\partial F}{\partial x} = y + 2\lambda x = 0 \Rightarrow y = -2\lambda x \\ \frac{\partial F}{\partial y} = x + 16\lambda y = 0 \Rightarrow x - 32\lambda^2 x = 0 \Rightarrow x(1 - 32\lambda^2) = 0 \\ g(x,y) = x^2 + 8y^2 - 1 = 0 \end{cases}$$

I $x = 0 \Rightarrow y = 0$, dar $0^2 + 8 \cdot 0^2 - 1 \neq 0 \Rightarrow$ nu e pct.

II $1 - 32\lambda^2 = 0 \Rightarrow \lambda^2 = \frac{1}{32} \Rightarrow \lambda = \pm \frac{1}{4\sqrt{2}}$

@ $\lambda = \frac{1}{4\sqrt{2}} \Rightarrow y = -2 \cdot \frac{1}{4\sqrt{2}} x \Rightarrow y = \frac{-1}{2\sqrt{2}} x \Rightarrow y^2 = \frac{1}{8} x^2$

$x^2 + 8 \cdot \frac{1}{8} x^2 - 1 = 0 \Rightarrow 2x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{1}{\sqrt{2}}$

$M_1\left(\frac{1}{\sqrt{2}}; -\frac{1}{4}\right) \quad M_2\left(-\frac{1}{\sqrt{2}}; \frac{1}{4}\right)$

⑥ $\lambda = \frac{-1}{4\sqrt{2}} \Rightarrow y = -2 \cdot \frac{-1}{4\sqrt{2}} x = \frac{1}{2\sqrt{2}} x \Rightarrow y^2 = \frac{1}{8} x^2$

$x = \pm \frac{1}{\sqrt{2}}$

$M_3\left(\frac{1}{\sqrt{2}}; \frac{1}{4}\right) \quad M_4\left(-\frac{1}{\sqrt{2}}; -\frac{1}{4}\right)$

$f(0;0) = 0$

$f\left(\pm\frac{1}{\sqrt{2}}; \pm\frac{1}{4}\right) = \frac{1}{\sqrt{2}} \cdot \frac{1}{4} = \frac{1}{4\sqrt{2}} = \sup_K f$

$f\left(\pm\frac{1}{\sqrt{2}}; \mp\frac{1}{4}\right) = \frac{-1}{\sqrt{2}} \cdot \frac{1}{4} = -\frac{1}{4\sqrt{2}} = \inf_K f$